Milestone 2 Report

ClimateScope: Visualizing Global Weather Trends and Extreme Events

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1. Introduction

Milestone 2 focuses on performing statistical analysis and designing visualization strategies to uncover trends, relationships, and extreme weather patterns within the Global Weather Repository dataset. The objective is to derive meaningful climate insights and design an interactive dashboard layout for effective data exploration.

The cleaned and pre-processed dataset from Milestone 1 was used as the foundation for this analysis.

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2. Statistical Distribution Analysis

Descriptive statistics were generated to understand the distribution and variability of key weather and environmental variables.

Temperature Distribution

- Mean Temperature: ~21.53°C
- Minimum: -29.8°C
- Maximum: 49.2°C
- Most observations fall between 15°C and 30°C

The temperature distribution shows moderate variability with some extreme cold and heat conditions, reflecting global geographic coverage.

Humidity

- Mean: ~49%
- Widespread across different regions
- Indicates diverse climatic zones represented in dataset

Wind Speed

- Significant variability observed
- Extreme values identified using percentile-based analysis

Air Quality Indicators

Air pollution metrics (PM2.5, PM10, Ozone, etc.) show substantial dispersion, highlighting regional differences in atmospheric conditions.

Some placeholder values (e.g., -9999) were identified and treated as invalid measurements (replaced with NaN) to prevent distortion of statistical analysis.

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3. Correlation Analysis

A correlation heatmap was generated to examine relationships among numeric variables.

Key Observations:

- Strong positive correlation between:
  - Temperature and Feels-Like Temperature
  - Wind Speed and Gust Speed
- Moderate correlation between:
  - Humidity and Cloud Coverage
  - Air Quality PM2.5 and Air Quality Carbon Monoxide
- Weak correlation between:
  - UV Index and most atmospheric variables

These findings indicate logical environmental relationships and confirm internal consistency within the dataset.

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4. Seasonal Trends & Time-Series Analysis

Monthly averages were computed using time-series resampling.

A line chart was used to analyse seasonal patterns.

Observations:

- Noticeable cyclical variation in temperature.
- Warmer periods show peak values mid-year.
- Some fluctuations in humidity and wind speed suggest seasonal shifts.
- Air quality indicators show regional variation over time.

The use of monthly aggregation reduces noise and highlights broader climate trends.

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5. Extreme Weather Event Identification

Extreme events were defined using the **95th percentile threshold method**.

Extreme Heat Events:

Observations above the 95th percentile of temperature were classified as extreme heat events.

These represent approximately 5% of the dataset and indicate potential heatwave conditions.

Extreme Wind Events

High wind speed values above the 95th percentile were isolated to identify severe wind occurrences.

Air Quality Extremes

Elevated PM2.5 levels were identified as potential pollution spikes.

This percentile-based method provides a statistically consistent approach to defining extremes rather than arbitrary thresholds.

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6. Regional Comparison Analysis

Country-wise aggregation was performed to compare average weather conditions across regions.

Key Findings:

- Tropical countries exhibit higher mean temperatures.
- Temperate regions show moderate seasonal variability.
- Pollution indicators vary significantly between countries.
- Wind intensity and humidity levels differ across geographic zones.

A choropleth map was used to visualize global temperature distribution, effectively highlighting geographic patterns.

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7. Visualization Strategy & Justification

The following visualization techniques were selected based on analytical objectives:

Analytical Objective	Visualization Type	Justification
Distribution analysis	Histogram	Shows data spread & skewness
Correlation analysis	Heatmap	Highlights relationships clearly
Seasonal trends	Line chart	Ideal for time-series visualization
Regional comparison	Bar chart	Effective categorical comparison
Geographic distribution	Choropleth map	Best for spatial intensity mapping

The selected visualizations collectively provide both macro-level insights and detailed analytical exploration.

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8. Dashboard Wireframe Design

The proposed interactive dashboard layout includes:

Header:

Project title and description.

Filter Panel:

- Country selector
- Date range selector
- Weather variable selector

Visualization Sections:

1. Monthly Trend Line Chart
2. Global Choropleth Map
3. Correlation Heatmap
4. Extreme Event Scatter Plot
5. Country Comparison Bar Chart

This layout ensures logical analytical flow:

- Filter → Trend → Geographic View → Relationships → Extremes → Comparison

The design supports intuitive navigation and multi-level exploration.

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9. Key Insights Summary

- Global temperature distribution shows expected regional variation.
- Strong relationships exist between related atmospheric variables.
- Seasonal trends are visible in monthly aggregated data.
- Extreme weather events can be identified using percentile-based statistical methods.
- Significant cross-country differences highlight the importance of geographic analysis.

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10. Conclusion

Milestone 2 objectives were successfully achieved:

- ✓ Statistical distribution analysis performed
- ✓ Correlation relationships examined
- ✓ Seasonal patterns identified
- ✓ Extreme weather events detected
- ✓ Regional comparisons conducted
- ✓ Visualization strategy defined
- ✓ Dashboard wireframe designed

The dataset now provides a strong foundation for developing an interactive climate analytics dashboard in the next milestone.